

Boosting CO₂ Electroreduction to CH₄ via Tuning Neighboring Single-Copper Sites

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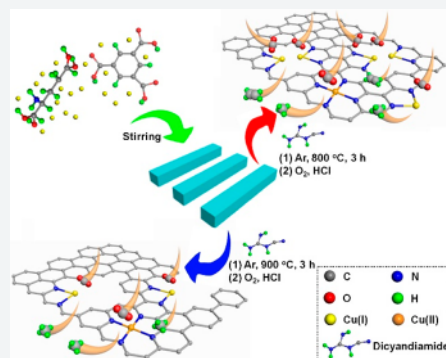


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Supporting Information

ABSTRACT: We report single-atom Cu catalysts dispersed on nitrogen-doped carbon by a nitrogen-coordination strategy. The presence of nitrogen enabled good dispersion and attachment of atomic Cu species on the nitrogen-doped carbon frameworks with Cu–N_x configurations. The Cu doping concentrations and Cu–N_x configurations were well-tuned by the pyrolysis temperature. At a high Cu concentration of 4.9%_{mol}, the distance between neighboring Cu–N_x species was close enough to enable C–C coupling and produce C₂H₄. In contrast, at Cu concentrations lower than 2.4%_{mol}, the distance between Cu–N_x species was large so that the electrocatalyst favored the formation of CH₄ as C₁ products. Density functional theory calculations further confirmed the capability of producing C₂H₄ by two CO intermediates binding on two adjacent Cu–N₂ sites, while the isolated Cu–N₄, the neighboring Cu–N₄, and the isolated Cu–N₂ sites led to formation of CH₄. Our work demonstrates a facile approach of tuning active Cu sites for CO₂ electroreduction to different hydrocarbons.



The greenhouse effect caused by excessive carbon dioxide (CO₂) emission into the atmosphere has been attracting increasing concern. The electrochemical conversion of CO₂ using sustainable energy sources is regarded as one of the promising approaches to reduce the continuous accumulation of atmospheric CO₂ and generate a supply of value-added chemical fuels to regain the carbon balance.^{1–3} Because of the intrinsic inertness of CO₂ molecules, the electrochemical CO₂ reduction reaction (CO₂RR) should rely on the development of optimal electrocatalysts that not only enable efficient CO₂ reduction under ambient conditions but also allow the production of deep reduction products beyond carbon monoxide or formic acid with good selectivity, such as methane (CH₄) and ethylene (C₂H₄).⁴ Among a wide range of material candidates that have been investigated for CO₂RR electrocatalysts, Cu-based materials are known as almost the only category that can electrochemically convert CO₂ to hydrocarbons and alcohols with good selectivities in aqueous solutions.^{5,6} Substantial research efforts have been focused on preparing Cu-based CO₂RR electrocatalysts from different aspects, including morphology,⁷ crystal plane,⁸ material composition,⁹ alloying,¹⁰ as well as the local pH near the electrode surface.¹¹

Methane, a common raw material for natural gas and a substitute for gasoline, is the deepest CO₂ reduction product for the C₁ pathway.¹² As it involves the transfer of 8 electrons and 8 protons, the electroreduction of CO₂ toward CH₄ is

substantially challenging. For instance, Yang and co-workers synthesized copper nanowires for CO₂RR and found the selectivity was sensitive to the nanowire morphological changes, among which the 5-fold twinned copper nanowires exhibited the best selectivity to CH₄.⁴ Wang et al. demonstrated that Cu atoms supported by CeO₂ nanorods could provide multiple oxygen vacancies to facilitate the electroreduction of CO₂ to CH₄.¹³ Despite these developments, Cu atoms/ions are generally prone to oxidation or agglomeration even under atmospheric conditions, resulting in both low catalyst stability and selectivity degrading.¹⁴ Recently, single-atom catalysts have been investigated as heterogeneous catalysts because of their unique electronic properties and maximized atomic utilization.¹⁵ For instance, many studies have shown that single-atom metal-based nitrogen-doped carbon (i.e., M–N–C, such as Co/Ni/Fe and Mn–N–C) catalysts are attractive CO₂RR electrocatalysts.^{16–24} Using a single-atom Ni–N–C catalyst, the maximum Faradaic efficiency (FE) of electroreducing CO₂ to CO was reported to be as high as ~99%.¹⁷ In the past few years, several Cu–N–C electrocatalysts^{25–27} were also reported to electrochemically

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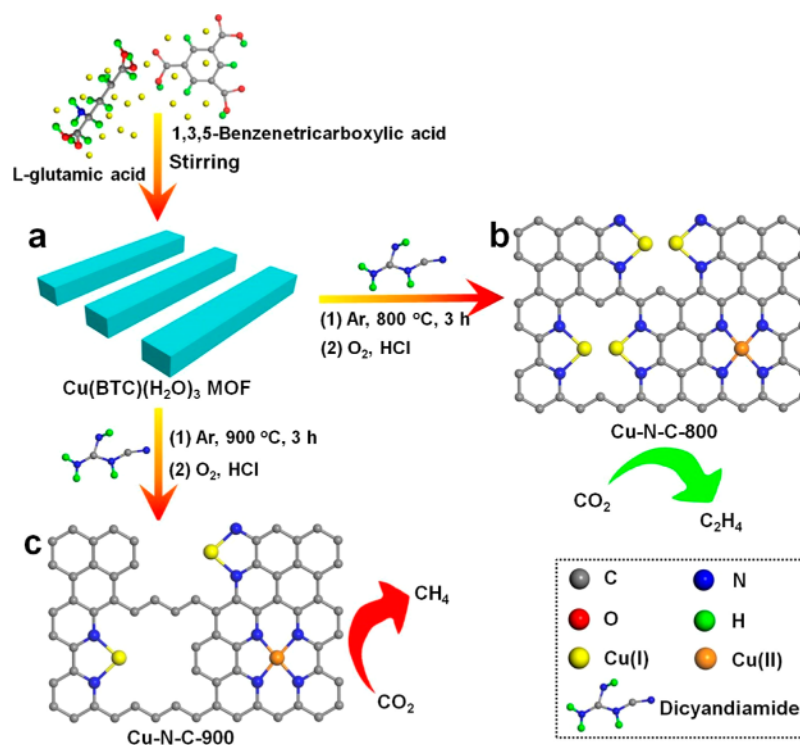


Figure 1. Schematic of the procedure to synthesize the Cu-N-C-T catalysts. (a) Cu(BTC)(H₂O)₃ MOF precursor. (b) Cu-N-C-800 synthesized by pyrolysis of Cu(BTC)(H₂O)₃ MOF and dicyandiamide at 800 °C, which favors the CO₂RR toward C₂H₄. (c) Cu-N-C-900 derived from the pyrolysis of Cu(BTC)(H₂O)₃ MOF and dicyandiamide at 900 °C, which favors the CO₂RR toward CH₄.

reduce CO₂ into CO at relatively less negative potentials. Nevertheless, the M-N-C electrocatalysts can only reduce CO₂ via the two-electron reduction products of CO and formic acid. Because the key step of producing deep reduction products, such as CH₄, is the generation of the *CHO intermediate by protonation of *CO, this step needs to overcome a large energy barrier. Therefore, CH₄ is preferentially produced at higher negative potentials.^{28,29} To date, there are few studies on the further reduction of CO₂ to hydrocarbons or alcohols catalyzed by single-atom M-N-C electrocatalysts.

In this work, we developed a Cu,N-codoped carbon nanosheet structure by atomically constructing the single-atom distribution of Cu-N_x coordination, with varying Cu-doping concentrations. The obtained single-atom Cu-N_x coordination structure enabled an effective CO₂RR catalytic activity toward hydrocarbons, including CH₄ and C₂H₄. At a high Cu loading density (4.9%_{mol}), Cu atoms were close enough to each other to enhance the C-C coupling, thus resulting in ethylene as a main CO₂ reduction product. When the Cu loading density was reduced to 2.4%_{mol}, the C₁ pathway was then favored and the main CO₂ reduction product was tuned to methane, thus suggesting a tailorable approach for constructing Cu-doped carbon framework for different CO₂RR selectivities.

The fabrication of Cu,N-codoped carbon nanosheets is schematically illustrated in Figure 1. First, 1,3,5-benzenetricarboxylic acid, copper(II) acetate monohydrate, and L-glutamic acid were used as original feedstock to synthesize a Cu(BTC)(H₂O)₃ metal-organic framework (MOF, Experimental Section in the Supporting Information). The crystal phase and morphology of the as-synthesized MOF were investigated by X-ray diffraction (XRD, Figure S1a), Fourier

transform infrared spectroscopy (FT-IR, Figure S1b), and scanning electron microscopy (SEM, Figure S2). Afterward, the Cu(BTC)(H₂O)₃ MOF was mixed with dicyandiamide (NH₂C(=NH)NHCN) and then calcined at high temperatures, during which the organic ligands were pyrolyzed to form N-doped carbon framework with Cu. Finally, the excessive Cu nanoparticles were etched by acid, while the nitrogen-coordinated Cu atoms remained in the carbon framework. The obtained products were designated as Cu-N-C-T (where T = 800, 900, 1000, and 1100 °C, indicating the calcination temperature). For comparison, Cu(BTC)(H₂O)₃ was also calcined at 900 °C without adding dicyandiamide to study the role of dicyandiamide in the formation of the Cu-N_x configuration in the obtained catalyst, and the resulting sample after acid leaching was designated as Cu-C-900.

The XRD patterns of all the Cu-N-C-T-BL samples (where BL means before acid leaching) exhibited a series of sharp diffraction peaks at 2θ of 43.3°, 50.4°, 74.1°, 89.9°, and 95.1° (Figures 2a and S3), corresponding to the (111), (200), (220), (311), and (222) crystal planes of the cubic Cu structure (JCPDS 040836). After the acid leaching, the obtained Cu-N-C-T samples showed only a broad peak centered at 26.4° for the (002) plane of the hexagonal graphitic structure (Figures 2a and S4).³⁰ The XRD results suggested that all the metallic state of Cu was removed by the acid leaching, as the Cu particles had weak interaction with the carbon matrix and were dissolved in the acid solution. For comparison, the XRD patterns of Cu-C-900-BL, Cu-C-900, and nitrogen-doped carbon (N-C) samples are also displayed (Figure S5), in which the diffraction peaks of Cu and carbon can be easily identified.

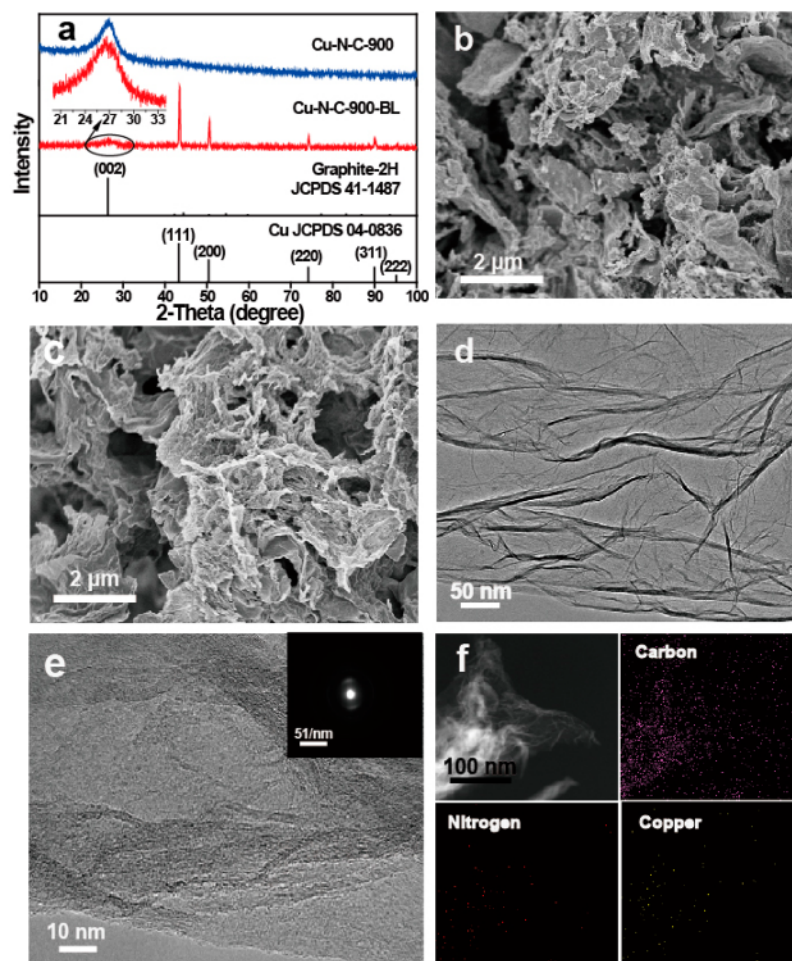


Figure 2. (a) XRD patterns of Cu-N-C-900-BL (red curve) and Cu-N-C-900 (blue curve) samples. SEM images of (b) Cu-N-C-900-BL and (c) Cu-N-C-900. (d) Low-resolution and (e) high-resolution TEM images of Cu-N-C-900. Inset in panel e: corresponding SAED pattern. (f) EDS elemental mapping of the Cu-N-C-900 sample.

Scanning electron microscopy images showed that both the Cu-N-C-900-BL and Cu-N-C-900 samples consist of interconnected foam structures with wrinkled and entangled flakes (Figure 2b,c). The low-resolution TEM image of Cu-N-C-900 catalyst presented cross-linked and ultrathin folded sheet-like features (Figure 2d). The high-resolution TEM image confirmed that there were no observable copper particles in the Cu-N-C-900 catalyst (Figure 2e), and the selected-area electron diffraction (SAED) pattern suggested the noncrystallinity of the Cu-N-C-900 sample (Figure 2e, inset). The high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) image also showed that no metallic copper aggregation on the ultrathin carbon nanosheets (Figure S6). The EDS elemental mapping revealed the uniform dispersion of the N and Cu atoms (Figure 2f).

The role of nitrogen doping was studied by comparison with samples undoped with N. All Cu-N-C-*T*-BL and Cu-N-C-*T* catalysts showed similar morphology to those of the N-C samples (Figures S7–S9), without obvious copper particles on the surface of carbon sheets. While without adding nitrogen, the Cu-C-900-BL sample retained the morphology of the starting material Cu(BTC)(H₂O)₃ MOF with lots of copper microspheres on the surface (Figure S10), which disappeared after treatment with oxygen-saturated acid (Figure S11). The TEM, elemental mapping (Figure S12), and HAADF-STEM

images (Figure S13) of all Cu-N-C-*T* samples were also used for further confirmation. All these results indicated that N atoms played a crucial role in stabilizing Cu atoms and prevented them from agglomeration during the high-temperature pyrolysis process, leading to a uniform distribution of Cu single atoms in the carbon matrix.

The N₂ sorption isotherms and pore diameter distributions demonstrated that all Cu-N-C-*T* samples possessed similar Brumauer–Emmett–Teller (BET) surface areas and pore volumes (Figures 3a and S14). For example, the Cu-N-C-900 exhibited a high BET specific surface area of 456.6 m²·g^{−1}, with a total pore volume of 3.05 cm³·g^{−1}, which could facilitate the transport of both CO₂ and electrolyte in the triple-phase contact area during the CO₂RR. Raman analysis was further conducted to evaluate the structures on these catalysts. As a representative, the Raman spectrum of Cu-N-C-900 was deconvoluted into four bands (Figure 3b). The first band peaked at *ca.* 1200 cm^{−1} is ascribed to polyene-like structures with sp³ phase (D₄-band),³¹ and the second band at *ca.* 1360 cm^{−1} is related to defects and dangling bonds in the carbon structures (D-band).³² The third band at *ca.* 1500 cm^{−1} corresponds to the amorphous sp² phase or/and Cu-N_x bond in carbon materials (D₃-band).³¹ The final band at *ca.* 1580 cm^{−1} is ascribed to the in-plane vibration of carbon atoms in the sp² electronic configuration (G-band).^{33–35} The

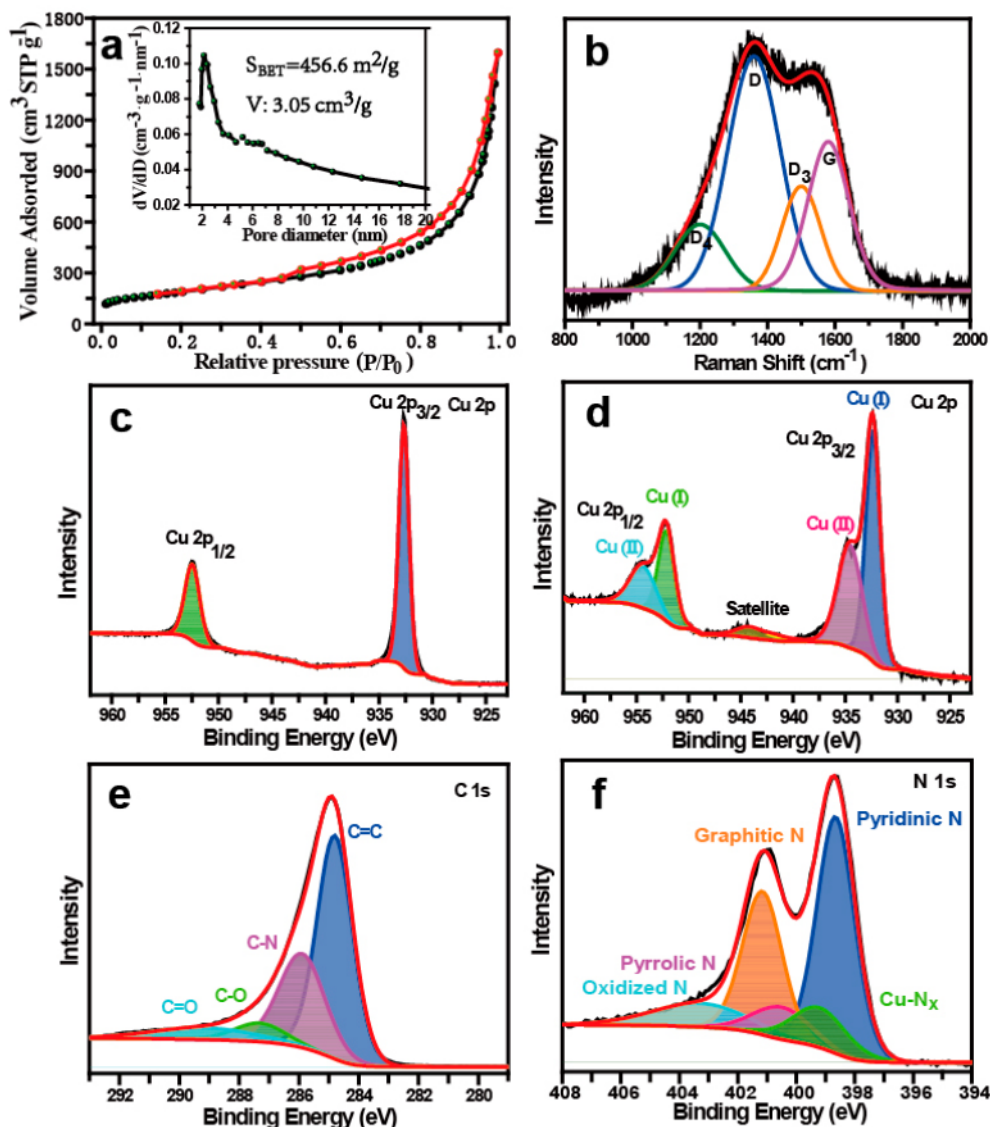


Figure 3. (a) N₂ sorption isotherms curve and (inset) pore size distribution for Cu-N-C-900. (b) Raman spectrum of Cu-N-C-900. (c) Cu 2p XPS spectrum of Cu-C-900-BL. (d–f) High-resolution XPS spectra of Cu-N-C-900 in the (d) Cu 2p, (e) C 1s, and (f) N 1s regions.

FT-IR spectra (Figure S15) did not present any obvious functional groups on all Cu-N-C-T catalysts, because of the high-temperature pyrolysis process during the sample synthesis, suggesting that surface functional groups had negligible influence on the catalytic performance of the Cu-N-C-T catalysts.

The chemical states of C, N, and Cu species of the Cu-C-900-BL and Cu-N-C-900 samples were studied using X-ray photoelectron spectroscopy (XPS). Three peaks were observed at 284.6, 286.8, and 289.1 eV in the high-resolution C 1s spectrum of the Cu-C-900-BL sample (Figure S16), attributed to the C=C (sp²-bonded), C–O, and C=O groups, respectively.³⁶ No signal of N element was detected in the Cu-C-900-BL sample (Figure S17). In the high-resolution Cu 2p region of the Cu-C-900-BL sample (Figure 3c), two peaks were observed at 932.3 and 952.5 eV, indicating that only the metallic state of Cu was found for the Cu-C-900-BL sample.³⁷ However, in the Cu 2p spectrum of the Cu-N-C-900 (Figure 3d), two main peaks centered at 932.4 and 952.1 eV were assigned to Cu(I)/Cu(0) species with a satellite peak at 942

eV; the peaks at 934.9 and 954.7 eV, as well as a satellite peak at 944.3 eV, were attributed to Cu(II) species with an unfilled electron state of the Cu 3d⁹ orbital.^{38,39} A comparison of the Cu 2p region between Cu-C-900-BL and Cu-N-C-900 samples (Figure S18) shows that the binding energy of Cu in Cu-N-C-900 shifted to a lower-energy region compared to that of Cu-C-900-BL, indicating that Cu(I) rather than Cu(0) was present in Cu-N-C-900.⁴⁰ In addition, the high-resolution C 1s spectrum of Cu-N-C-900 was fitted into four peaks centered at 284.6, 285.9, 287.2, and 289.1 eV, corresponding to C=C (sp²-bonded), C–N (sp³-bonded), C–O, and C=O groups, respectively (Figure 3e).³⁶ The deconvolution of the N 1s peak showed four peaks that were assigned to pyridinic (398.7 eV), pyrrolic (400.5 eV), graphitic (401.1 eV), and oxidized N (403.3 eV) species (Figure 3f).^{41–43} Additionally, the peak positioned at 399.2 eV indicated the coordination of N atoms with Cu, supporting the existence of Cu–N bonds in the Cu-N-C-900 sample.^{18,27} The XPS spectra of the other samples are also displayed (Figures S19 and S20) for comparison, all of which

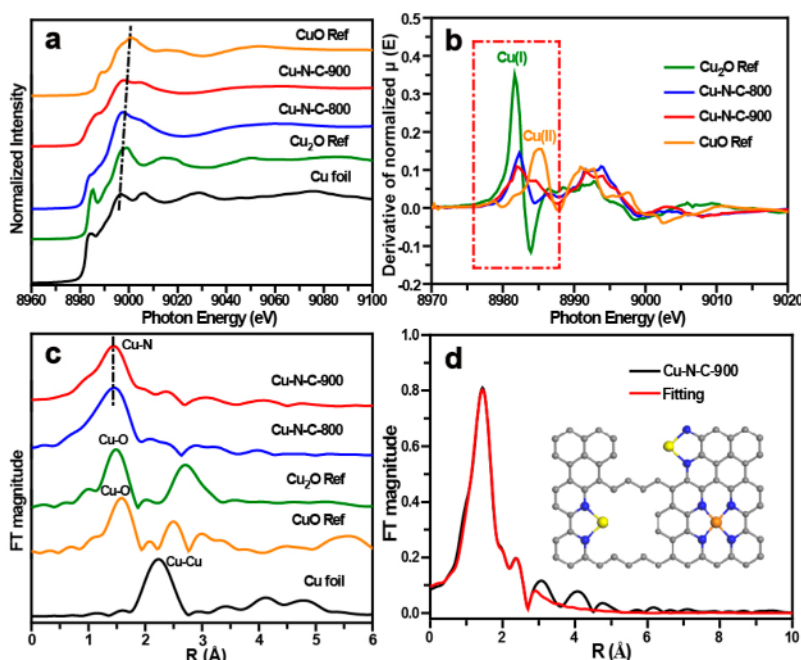


Figure 4. (a) XANES spectra of Cu–K edge for Cu foil reference, Cu₂O reference, CuO reference, Cu–N–C-800, and the Cu–N–C-900 samples. (b) First derivatives of the XANES region for Cu₂O, CuO, Cu–N–C-800, and the Cu–N–C-900 samples. (c) FT-EXAFS spectra of Cu–K edge for Cu foil reference, Cu₂O reference, CuO reference, Cu–N–C-800, and Cu–N–C-900 samples. (d) EXAFS R-space fitting curves of Cu–N–C-900 catalyst. Inset: illustration for the Cu–N₂ and Cu–N₄ coordination structures. Gray, blue, yellow, and orange spheres represent C, N, Cu(I), and Cu(II) elements, respectively.

show features similar to that of the Cu–N–C-900 sample. The analysis of the relative amount of the pyridinic and graphitic N species in the Cu–N–C-*T* samples revealed that the amount of pyridinic N decreased with the increase of the pyrolysis temperature, while the graphitic N species showed a gradual increasing trend. These trends suggest a gradual thermal transformation of pyridinic N to graphitic N with the increasing temperature, in agreement with the literature that higher temperature leads to the formation of graphitic N species.^{44,45}

The X-ray absorption of Cu K-edge was conducted to investigate the valence and coordination states of Cu in the Cu–N–C-800 and Cu–N–C-900 catalysts (Figure 4a). Compared to the reference sample of Cu foil with a characteristic peak at ~8995 eV, the main peak of the Cu–N–C-900 sample shifted to a higher-energy region, suggesting that the Cu atoms were in higher oxidation states of Cu(I) or Cu(II).^{35,46} Moreover, the characteristic peaks of the Cu–N–C-800 and Cu–N–C-900 samples in the first derivative spectrum existed in the regions between Cu(I) and Cu(II) (Figure 4b), further confirming the Cu atoms were in the oxidation states.⁴⁷ The atomic structure of the sample was investigated by the Fourier-transformed (FT) *k*³-weighted extended X-ray absorption fine structure (FT-EXAFS). As shown in Figure 4c, the presence of a peak located at 1.45 Å was attributed to the Cu–N bond, and the absence of the peak at around 2.24 Å (i.e., Cu–Cu bond) suggested that the Cu atoms were separated from each other and coordinated with N atoms,³⁸ which is similar to the XPS result. The EXAFS spectra were further fitted to obtain the structural configuration of Cu in the Cu–N–C-800 and Cu–N–C-900 samples (Figures 4d, S21, and 22), and the fitting parameters are listed in Table S1. It could be seen that Cu atoms were coordinated with nitrogen atoms, and the scattering path of (CuN₂)–(CuN₂) in the Cu–

N–C-800 sample was much smaller than that of the Cu–N–C-900 sample. This significantly reduced distance value suggests that a synergistic effect can occur between the two neighboring Cu–N₂ sites during the CO₂RR. According to the EXAFS and the Cu 2p XPS results, it could be concluded that the Cu coordination structure with two different valence states existed in the Cu–N–C-800 and Cu–N–C-900 catalysts, namely, Cu–N₂ and Cu–N₄.³⁵ On the one hand, one Cu ion and two pyridinic-N atoms form the Cu–N₂ species with Cu(I) ion as the oxidized species. On the other hand, the Cu–N₄ structure is formed by the symmetric coordination of one Cu ion with four pyridinic-N atoms in the inner region of carbon frameworks and the Cu(II) ion is the oxidized species.^{35,38} Based on the above analysis, the local atomic structure around Cu can be constructed. The Cu–N–C-800 sample possesses neighboring Cu–N₂ sites as well as isolated Cu–N₄ sites (Figure S21, inset), in comparison to the isolated Cu–N₂ and Cu–N₄ structures of the Cu–N–C-900 sample (Figure 4d, inset). These Cu–N_x chemical structures can serve as excellent electrocatalytic active sites for the CO₂RR.

The electrochemical CO₂ reduction reaction was carried out in a commercial H-type cell with a cathodic compartment and an anodic compartment separated by a Nafion membrane (Figure S23). The electrochemical performances of these materials were first investigated by linear sweep voltammetry (LSV) in 0.1 M KHCO₃ solution (Experimental Section in the Supporting Information). Compared to the HER with the purging of Ar gas (pH 8.3), the Cu–N–C-900 catalyst in the CO₂-saturated electrolyte (pH 6.8) exhibited a much higher current density for the CO₂RR (Figure S24), indicating its excellent catalytic performance for CO₂ electroreduction. Moreover, it presented the smallest (negative) onset potential at –0.7 V vs RHE and the largest reduction current density (Figure 5a, red curve) in comparison to those of Cu–C-900

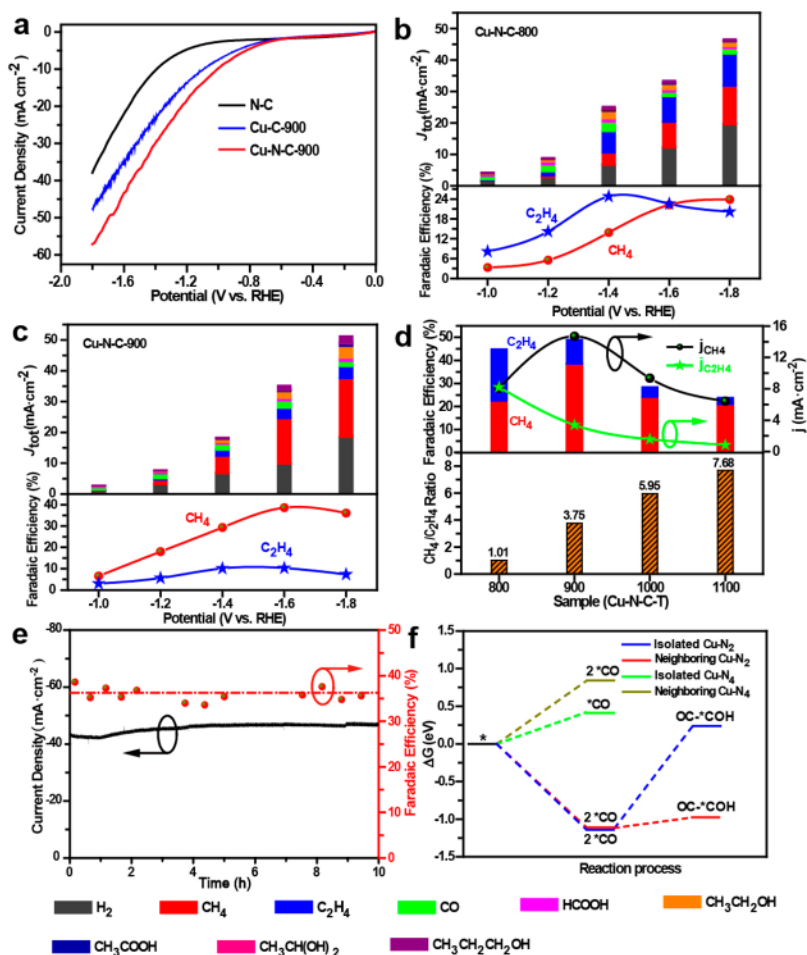


Figure 5. (a) Linear sweep voltammetry curves for the N-C, Cu-C-900, and Cu-N-C-900 catalysts. (b) CO₂ electroreduction performance of the Cu-N-C-800 catalyst. Top panel: current densities, with colors indicating different products as shown at the bottom. Bottom panel: Faradaic efficiencies of C₂H₄ and CH₄. (c) CO₂ electroreduction performance of the Cu-N-C-900 catalyst. (d) Faradaic efficiencies (top panel, left y-axis), partial current densities of CH₄ and C₂H₄ (top panel, right y-axis), and the ratios of CH₄/C₂H₄ (bottom panel) for Cu-N-C-T (T = 800, 900, 1000, and 1100 °C) catalysts at −1.6 V vs RHE. (e) Stability of Cu-N-C-900 for electrochemical reduction of CO₂. Left y-axis, total current density (black curve); right y-axis, Faradaic efficiencies of forming CH₄ (red dots). (f) Calculated Gibbs free-energy diagrams for CO₂ electroreduction on different Cu-N_x coordination structures.

(blue curve) and the N-C samples (black curve). The in-line gas chromatography (GC) and ¹H nuclear magnetic resonance (¹H NMR) measurements were used to quantify the electroreduction products of these catalysts. Both the Cu-C-900 (Figure S25a) and N-C catalysts (Figure S25b) showed negligible catalytic activities for the CO₂RR and displayed almost all Faradaic efficiencies toward the HER at various applied potentials, indicating that oxygen and other elements derived from Cu(BTC)(H₂O)₃ MOF had negligible influence on the CO₂RR performance. In contrast, a distinctively different catalytic selectivity was observed for the Cu-doped N-C samples. The current densities and corresponding Faradaic efficiencies for H₂, CH₄, C₂H₄, and other products are shown for Cu-N-C-800 (Figure 5b) and Cu-N-C-900 catalysts (Figure 5c), as a function of the cell voltage. As observed, the Cu-N-C-800 catalyst exhibited an appreciable FE_{C₂H₄} (24.8%) for C₂H₄ production at −1.4 V vs RHE (Figure 5b), superior to its CH₄ formation with a FE_{CH₄} of 13.9%. The partial current densities for C₂H₄ and CH₄ formation were 6.84 and 3.83 mA·cm^{−2} at −1.4 V vs RHE, respectively. On the other hand, for the Cu-N-C-900 catalyst, the C₂H₄

production was largely suppressed, and the CO₂RR products were mainly composed of CH₄ at the same applied potentials (Figure S26). The highest FE_{CH₄} production reached 38.6% at −1.6 V vs RHE, corresponding to a partial current density for CH₄ production of 14.8 mA·cm^{−2} (Figure 5c).

To better evaluate the effect of pyrolysis temperature on the CO₂ electroreduction selectivity, the Cu-N-C-1000 and Cu-N-C-1100 catalysts were also tested for their CO₂RR activities and compared. As shown in Figures S27 and 5b,c, the Cu-N-C-800 sample clearly showed high CO₂RR selectivity to C₂H₄, while the other catalysts tended to produce CH₄ as the main product and suppress C₂H₄. Figure 5d displays the Faradaic efficiencies, partial current densities of CH₄ and C₂H₄ for all Cu-N-C-T samples, and the ratios of CH₄ and C₂H₄ at −1.6 V vs RHE. The ratios of CH₄/C₂H₄ were tuned from 1.01 to 7.68, indicating the formation of CH₄ was favorable with the increase of pyrolysis temperature during the catalyst synthesis process. However, the Faradaic efficiencies of the other products, such as ethanol, for all Cu-N-C-T samples were all below 10% (Figure S28), indicating that the pyrolysis temperature had little effect on the formation of ethanol or other products.

Catalyst stability during the CO₂RR reaction is an important factor for industrial application, which can be influenced by the pH change of electrolyte,^{11,48} the deposition of impurities from the solution,⁴⁹ the structure change of the catalyst, and so on.⁵⁰ Therefore, the stability of the Cu–N–C-900 catalyst was studied under the electrochemical reduction reaction. The current density showed no obvious attenuation after 10 h of continuous electrolysis at –1.6 V vs RHE (Figure S5e). The detection of gas products confirmed that the Cu–N–C-900 catalyst maintained a stable production rate for CH₄ formation (FE ≈ 35.6% after 10 h of electrolysis) during this process. After the CO₂RR, the XPS spectra of all the Cu–N–C-*T* samples (Figure S29) revealed that the N species and the valence state of Cu sites remained almost unchanged, indicating their robustness during the CO₂RR process. Moreover, the XAS measurement was further conducted to probe the structural and oxidation states of the Cu–N_x configurations after the CO₂ reduction reaction. The Cu K-edge XANES spectra (Figure S30a) of the Cu–N–C-800 and Cu–N–C-900 catalysts after CO₂RR verified that Cu atoms in these samples were still in the oxidation states of Cu(I) and Cu(II), similar to those before the CO₂RR. The corresponding EXAFS spectra (Figure S30b) after the CO₂RR also demonstrated that Cu atoms existed as Cu–N_x species in the samples with the characteristic peak at 1.45 Å, and without any Cu–Cu peak at around 2.24 Å,³⁸ further confirming the excellent stability of these catalysts toward the electrochemical reduction of CO₂.

A mechanism was then proposed to illustrate why the selectivity of CO₂ electroreduction varied with the pyrolysis temperature of catalysts. First, as shown in Table S2, all the Cu–N–C-*T* samples with no obvious functional groups on the surface exhibited the same morphological features and similar BET surface areas, indicating that these parameters were not the critical factors that influenced the selectivity of the CO₂RR. However, the Cu contents for all catalysts were measured (Table S3), and the results showed that the contents of Cu decreased gradually with the increase of temperature. When calcined at high temperature, some copper atoms coordinated with nitrogen atoms were remained in the carbon frameworks with Cu–N_x coordination structure, while the other copper atoms were detached from the carbon matrix and converted into metallic copper. With the further increase of calcination temperature, more and more copper atoms were removed from the carbon networks and aggregated into nanoparticles, along with the evaporation of nitrogen. As shown in Table S4, the N concentration gradually decreased with increasing pyrolysis temperature for the Cu–N–C-*T* series, and the Cu–N–C-900 catalyst possessed a similar N concentration to that of the N–C sample because both of them were pyrolyzed at the same temperature. The Cu–C-900 catalyst showed the lowest N concentration because of a lack of external nitrogen source during its synthesis procedure. These results suggested the successful doping of nitrogen into the carbon frameworks, and the nitrogen concentration decreased at high temperatures because of its volatilization.

As when a *CO intermediate is produced on one Cu site, it either desorbs to form CH₄ by following the pathway *CO → *CHO → *CH₂O → *CH₃O → CH₄ or couples with another *CO intermediate on the neighboring single-atom Cu site to form C₂H₄ through a C–C bond-formation step.⁵¹ The Cu–N–C-800 catalyst shows a more favorable selectivity toward C₂H₄ formation than that of CH₄, which is ascribed to the

existence of abundant single-atom Cu active sites and the synergistic effect between them to induce the C–C coupling (Figure S31). At pyrolysis temperatures higher than 800 °C, more and more atomic Cu species detach from the carbon frameworks, and the remaining single-atom Cu species are isolated from each other, leading to the increase of CH₄ products and the decrease of C₂H₄ products (Figure S32). However, the more drastic pyrolysis treatment at a higher temperature (1100 °C) resulted in the deterioration of CO₂RR activity and enhancement of the HER side reaction, as a vast majority of active sites (Cu single atoms) for CO₂RR were dissociated from the carbon framework at high temperatures. Therefore, more and more naked carbon framework without copper single atoms was obtained and thus became an inert catalyst for CO₂ electroreduction but showed increased HER activity.⁵² Thus, it can be concluded that the catalytic selectivities of our catalysts are strongly related to the contents of monodispersed Cu atoms on nitrogen-doped carbon with the Cu–N_x configuration. In our work, when the Cu concentration reaches ~4.9%_{mol}, the Cu single atoms are close enough to each other, which leads to the presence of a synergistic effect between adjacent Cu single atoms and favors the dimerization of *CO intermediates. At a lower Cu concentration below 2.4%_{mol}, Cu single atoms are isolated sufficiently far from each other, thus resulting in low surface coverage of *CO species and enhancing the formation of CH₄.

To further confirm our proposed mechanism, density functional theory (DFT) calculations were performed to demonstrate the product selectivity of CO₂RR on different Cu–N_x coordination structures (Figures S5f and S33). The reaction procedures on the isolated and neighboring Cu–N₄ are both thermodynamically uphill (Figure S5f), suggesting inefficient electrochemical CO₂RR performance. The gentle upslopes belonging to the CO binding strengths on the isolated Cu–N₄ and the neighboring Cu–N₄ are both weak and unable to favor the C–C coupling, suggesting that these two sites prefer to produce C₁ products. On the other hand, the CO binding strengths on the isolated Cu–N₂ (–2.36/2 = –1.18 eV) and the neighboring Cu–N₂ (–2.34/2 = –1.17 eV) are both obviously stronger than that on the isolated Cu–N₄ (–0.20 eV) and the neighboring Cu–N₄ sites (–0.38/2 = –0.19 eV), giving a strong indication that these two sites can facilitate further CO reduction. In addition, OC–*COH (* denotes the binding site) is known to be a common intermediate proposed in the literature for the formation of C₂ or C₂₊ products.^{53,54} As shown in Figure S33i,j, the free energy of C–C coupling on the isolated Cu–N₂ is as high as 1.38 eV, while that on the neighboring Cu–N₂ is only 0.14 eV. This is consistent with our experimental observation that a high Cu concentration site enables the C–C coupling to produce C₂H₄, whereas a low Cu concentration site is unable to play such a role.

In summary, we have demonstrated a temperature-tuned N-coordination strategy to disperse a high content single-atomic Cu on nitrogen-doped carbon for CO₂ electroreduction. Our results show that nitrogen could anchor single-atomic copper species (up to ~4.9%_{mol}) on the carbon frameworks with the Cu–N_x configuration, which was tailored by altering the calcination temperature. These Cu–N_x species were confirmed to be the main active sites for CO₂RR. In addition, the neighboring single-atomic Cu sites were in favor of C₂H₄ formation, while isolated single-atomic Cu functioned as active sites for CH₄ production. Thus, by tuning the pyrolysis

temperature and the obtained Cu–N_x species, the CO₂RR selectivity was tuned from C₂H₄ to CH₄, leading to a significant change of ratio between CH₄ and C₂H₄. DFT calculations further suggest that high Cu concentration with neighboring Cu–N₂ sites is more beneficial for the C₂H₄ generation because of the low free energy for C–C coupling, while the isolated Cu–N₄, the neighboring Cu–N₄, and the isolated Cu–N₂ are all inclined to form C₁ products. Our work suggests the importance of tuning the distance between single Cu atomic sites for CO₂RR, which should be thoroughly considered in designing the CO₂RR to hydrocarbons.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsenerylett.0c00018>.

Experimental details and data (Figures S1–S33 and Tables S1–S4) (PDF)

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Notes

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